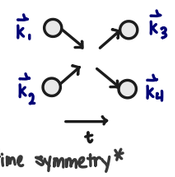
Transition Rates

$\Gamma_{i\to f} \propto (1+N_f)$ bosons $\quad \int N(E)\,D(E)\,dE = N_{TOT}$

$\Gamma_{i\to f} \propto (1-N_f)$ fermions $\quad \int N(E)\,D(E)\,dE = U_{TOT}$

density of states

Boltzmann's argument of "detailed balance"



time symmetry*

Forward Scattering Rate

$$\Gamma_{i\to f} \propto (M_{1234})^2 N_1 N_2 (1\pm N_3)(1+N_4)$$

QM terms

'Matrix Element' — numbers in states K_1 and K_2

Reverse Process

$$\Gamma_{i\to f} \propto (M_{3412})^2 N_3 N_4 (1\pm N_1)(1+N_2)$$

by time-reversal symmetry: $M_{1234} = M_{3412}$

* detailed balance
↳ in equilibrium
rate forward = reverse role

$$\cancel{(M_{1234})^2} N_1 N_2 (1\pm N_3)(1+N_4) = \cancel{(M_{3412})^2} N_3 N_4 (1\pm N_1)(1+N_2)$$

$$N_1 N_2 (1\pm N_3)(1+N_4) = N_3 N_4 (1\pm N_1)(1+N_2)$$

Guess sol$^{\underline{n}}$: $N(E) = \dfrac{1}{e^{\beta E + \alpha} \pm 1}$ (KE) — − bosons, + fermions

$$\left(\frac{1}{e^{\beta E_1+\alpha}\pm 1}\right)\left(\frac{1}{e^{\beta E_2+\alpha}\pm 1}\right)\underbrace{\left(\frac{e^{\beta E_3+\alpha}\cancel{\pm 1}\pm 1}{e^{\beta E_3+\alpha}\pm 1}\right)}_{1+N_3}\left(\frac{e^{\beta E_4+\alpha}\cancel{\pm 1}\pm 1}{e^{\beta E_4+\alpha}\pm 1}\right)$$

- $\alpha = \dfrac{-\mu}{k_B T}$
- $\beta = \dfrac{1}{k_B T}$

$$e^{\beta E_3+\alpha} e^{\beta E_4+\alpha} = e^{\beta E_1+\alpha} e^{\beta E_2+\alpha} \Rightarrow e^{\beta(E_3+E_4)}\cancel{e^{2\alpha}} = e^{\beta(E_1+E_2)}\cancel{e^{2\alpha}}$$

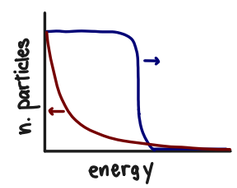
by law of conservation

* true if $E_1 + E_2 = E_3 + E_4$!

$$N(E) = \frac{1}{e^{(E-\mu)/k_BT} - 1}$$ "Bose-Einstein Distribution"

- μ = 'chemical potential'

$$N(E) = \frac{1}{e^{(E-\mu)/k_BT} + 1}$$ "Fermi-Dirac Distribution"



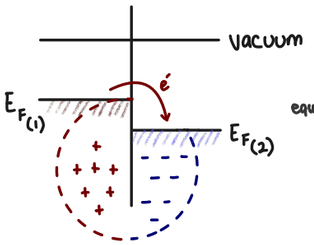
Bound States

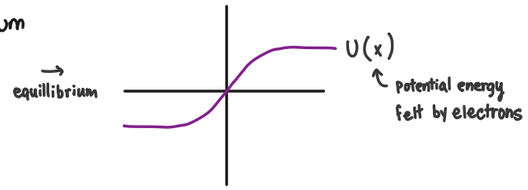
fermions $\pm\frac{1}{2}, \ldots$

bosons $0, \pm 1, \ldots$

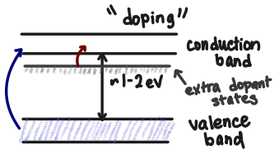
^4He: $2p^+$, $2e^-$, $2n^0$ — "superfluid" → 6 fermions

Cooper pair — 'superconductor' "condensate"
bound state of 2 fermions
spin: $\pm\frac{1}{2}$





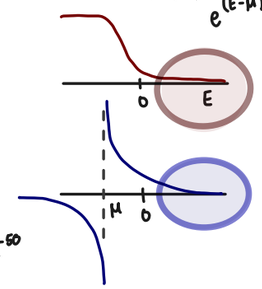
Diodes and Transistors

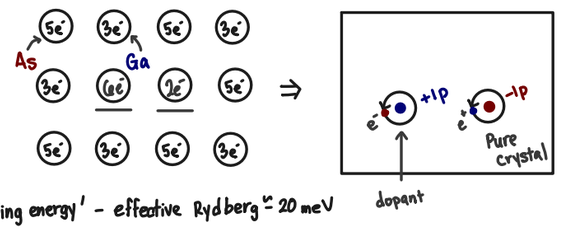


k_B - 'Boltzmann's constant'

$k_BT \simeq 25.6$ meV for T = 300K (room T)

$e^{-E/k_BT} \simeq e^{-(1eV)/(.0256eV)} \simeq e^{-50}$





'binding energy' - effective Rydberg $\simeq 20$ meV

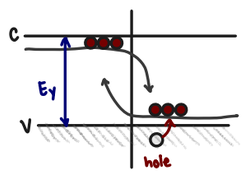
$$\frac{e^2}{r} \Rightarrow \frac{e^2}{\epsilon r}$$ (ϵ = dielectric constant)

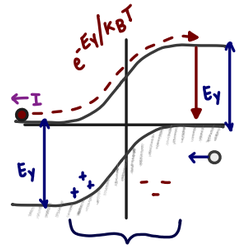
- dope w/ extra e^- 'donor'
  → donates free electrons n-type
- dope w/ missing e^- 'acceptor'
  → donates holes p-type

"majority" carrier - what sign is moving charge

p-n Junction

"diode"

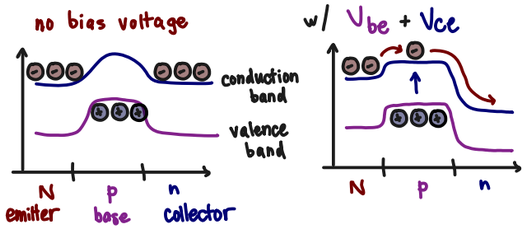


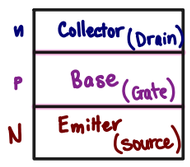


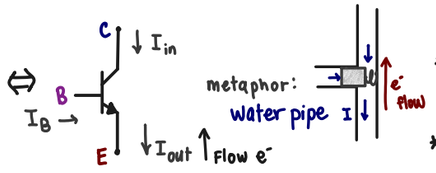
- electrons moving L→R feel potential energy barrier $(E_Y - q\Delta V)$ (applied voltage)

current: $I \propto e^{-(E_Y - q\Delta V)/k_BT} = e^{-E_Y/k_BT}\left(e^{q\Delta V/k_BT}\right)$

Bipolar Transistor



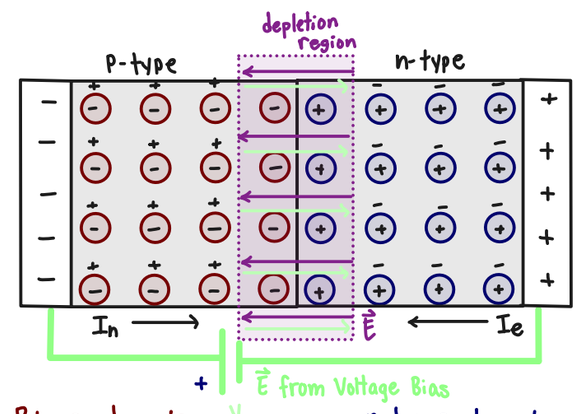




- If $V_{be} > 0.6V$ and $V_{ce} > V_{be}$
  ↳ then electrons can flow from E to B

* some e^- can recombine with holes in base creating I_b

* some e^- are pushed from B to C by applied V_{ce} and create I_c



P type dopants
- extra 'holes'

n type dopants
extra electrons

depletion region
- no extra carriers

Half-wave